1 Saving an activation adds a rounding step



No probe

q → **normalize + rotate** (one kernel, full precision) → *query*



Probe attached

q → **normalize** → BF16 → **rotate** → *query′*

query ≠ *query′*

last bits differ

2 One attention head reads the difference

Layer 35, head 6

<|im_start|> | user | Is a read-only probe … ? | assistant | ↵ → **Yes**

same offset in every prompt